

KALOL INSTITUTE OF TECHNOLOGY
IT Department
Assignment 1 TO 5
Subject : CNS(3161606)

WRITTEN BELOW 25 QUESTION AS A ASSIGNMENTS IN NOTEBOOK

1. Define:1). Cryptography 2). Cryptanalysis 3). Confidentiality 4). Data integrity.
2. List out various block cipher modes of operation and explain any 2 with the help of diagram.
3. Explain security Attacks.
4. Explain single round of DES algorithm.
5. Explain the key generation in AES algorithm.
6. Discuss RSA algorithm with diagram & suitable example.
7. Explain substitution and transposition techniques.
8. Briefly explain the logic of secure hash algorithm.
9. Difference between Symmetric and Asymmetric key cryptography.
10. Explain the key generation in AES algorithm..
11. Briefly explain Diffie-Hellman key exchange with diagram & example.
12. Explain different **types of security services** in detail.
13. Describe the **Euclidean Algorithm** with an example.
14. Compare **group, ring, and field** with suitable examples.
15. Explain **transposition techniques** with an example.
16. Differentiate between **stream cipher and block cipher**.
17. Write short notes on **design principles of block cipher**.
18. Explain **AES structure**.
19. Compare **DES and AES**.
20. Discuss **cryptography and cryptanalysis** along with types of attacks.
21. Explain **block cipher modes of operation**: 1)ECB 2) CBC 3) CFB 4) OFB 5) CTR
22. Describe **finite fields and their role in cryptography**.
23. Explain **modular arithmetic and Euclidean algorithm in cryptographic applications**.
24. Use the **Euclidean Algorithm** to find GCD of: (252, 105)
25. Solve using **modular arithmetic**: 1) $17 \bmod 517 \setminus \bmod 517 \bmod 5$ 2) $23 \bmod 723 \setminus \bmod 723 \bmod 7$